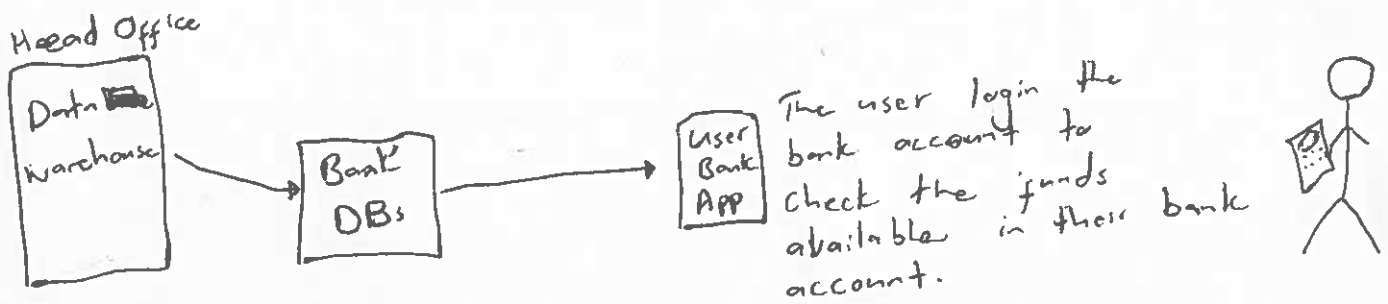


① Database (db) is where data/information of an organisation or institution is stored electronically in a computer system.

Database is essential because it can be fast to retrieve data, it secures data, allows many people to view and update information at exact same time without crashing or mixing the updated data, it built-in rules to make sure information stays accurate and reliable across all systems

examples (1) Banking - bank uses DBs to record accounts balance, process ATM withdrawals and log money transfers instantly and securely
(2) Sales Online retail - Takealot - use DBs to store product prices, track real-time stock levels and manage orders



② Data warehouse it is a centralized computer systems that collect, integrates and stores large volume of data current and historical from multiple database systems. It differs from regular DB primarily by purpose of analytical processing - it is designed for complex queries, forecasting and trend analysis. It uses OLAP (Online Analytical Processing) technology. It deals with historical data. Company use a data warehouse to make major analytical performance, to observe long-term trend analysis, to break down data silos and to employ high-quality business intelligence.

③ Data lake is a centralized, highly scalable repository that stores vast amounts of raw unstructured, semi-structured and structured data in its original form. This allows an organisations to store everything and deal with data later on.

Data warehouse stores filtered data, while a data lake stores massive, raw unfiltered data. Main use of data lake is for machine learning, predictive modeling, big data analytics and raw data archiving.

Business will prefer to use data lake when dealing with high-velocity 'big data' or advanced analytical applications examples: empowering machine learning and artificial intelligence - handling unstructure and diverse data types like images, music files, social media feeds, streams, server metrics. and real-time data streaming

④ Data mart is a focused, specialised subset of a data warehouse that contains data tailored specifically for a single department, line business or specific group of users. It relates to data warehouse by working together to organise an organisation data, usually interacting in one of two structural frameworks. Data marts are used by business end-users and departmental analysts e.g. Marketing teams, Finance & Accounting analysts, Sales managers and human resource personnel.

⑤ Structured data is organized, quantitative data that conforms to a strict format or data model of rows and columns, making it easily searchable by basic human queries and ~~standard~~ standard relational database algorithms.

Semi-Structure data does not conform to a rigid relational database structure and it is not completely random either. it contains internal tags, keys or organisational markers that separate data elements and enforce a visible hierarchy, making it flexible yet machine readable

Unstructured data is information that does not have conceptual structure or model. It is qualitative, heavily text-based or binary, and cannot be stored in traditional rows and columns without losing its meaning.

e.g.

Feature	Structured	Semi-structured	Unstructured
Format	Table (Rows/Columns)	Key-value pairs, Tags	Audio, Video, Free-text
Flexibility	Low (Strict schema)	Medium (Flexible schema)	High (No schema)
Storage Type	Data Warehouse / SQL DB	NoSQL DB / Data Lake	Data Lake / Object Storage

⑥ Data modeling is the process of creating a visual blueprint or conceptual diagram that defines how data is structured, stored and related within an information system.

It is important to prevent data redundancy and errors, to improve system performance, to align business goals with technical designs and reduces development costs.

Types

- ① Conceptual Data Model (Business View) - this is high-level, abstract model designed to define what the system will contain. It focuses entirely on identifying the core business entities and how they relate to one another.
- ② Logical Data Model (Structure View) - this model builds on conceptual model by adding more detail, defining how the system should be structured regardless of the specific database technology used.
- ③ Physical Data Model (Technical View) - this is the final, highly detailed blueprint that describes exactly how the data will be implemented using a specific database management system.

⑦ Data mining is a process of sorting through massive, raw ~~datasets~~ datasets to identify hidden patterns, norms, correlations and trends that can predict future behaviors or outcomes. It combines techniques from statistics, artificial intelligence and machine learning to turn overwhelming amounts of raw data into actionable knowledge.

Business uses: this for customer segmentation (grouping customers), churn prediction (identifying customer behavior), fraud detection, supply chain optimization

Two examples:

(1) Netflix - hyper-personalized recommendation engines - this uses sophisticated data mining algorithms to look at more than just what movie you clicked on. It mines data on when you watch, how long you watch, what device you use, when you pause and even how fast you scroll past certain titles.

(2) Target: Predictive target marketing - in a famous and foundational retail example, the retail giant Target used data mining to predict major life events of their shoppers.

⑧ Database Management System (DBMS) is a specialised software application that acts as an interface between the database and its end users / programs. It allows interaction for users to create, read, update, delete, and manage data systematically.

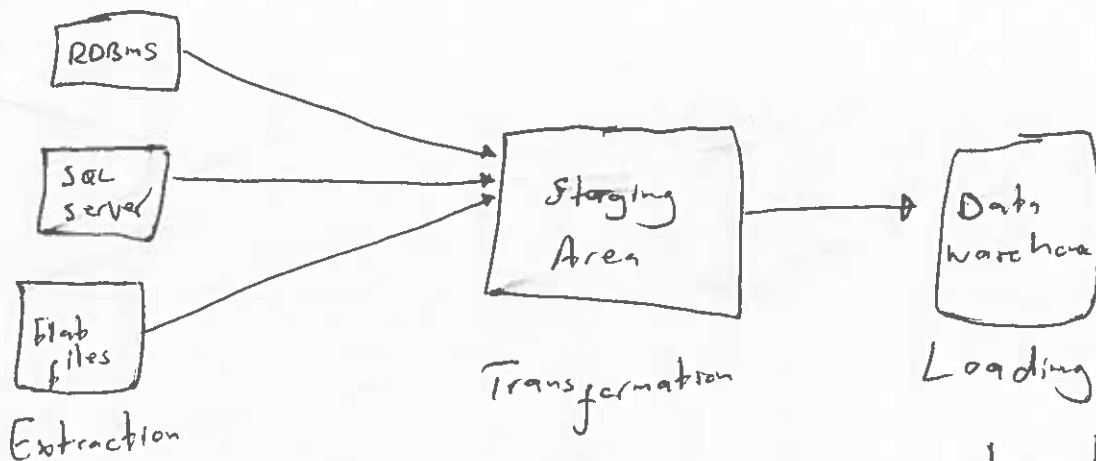
(1) PostgreSQL - it is an advanced, open-source object-relational database management system. Suitable use on financial and banking platforms.

(2) MongoDB - it is a popular, open-source NoSQL database that stores data as flexible, rather than rows and columns. Suitable use on content management systems or E-commerce product catalogs.

(3) MySQL - it is most popular, open-source relational database. Suitable use on web applications and content platforms.

(4) Oracle - it is an enterprise grade relational database management system designed for massive corporate deployment. Suitable use in large-scale enterprise resource planning.

9



ETL (Extract, Transform, Load) is a data integration process used to pull raw data from multiple places, clean and format it, then store it.

This process occurs in three steps

Extract - data comes from different sources such as database, customer relationship management (CRM) software, APIs or flat files and moved into a temporary staging area. This stops the process from slowing down live production systems.

Transform - the raw data is cleaned, checked for errors and changed into a standard format. This step removes duplicate entries, fixes missing values and applies business rules so the data matches the target system.

Load - the clean and organized data is written into a final destination, such as a data warehouse or data lake.

This process is important because it creates a single source of truth, improves data quality, speeds up queries and supports better decisions.

10

Format	Structure Type	Best for	Pros	Cons
CSV	Row-oriented (Flat)	Simple, small data exchange	Universal compatibility, human-readable	Slow for large datasets, no built-in schema
JSON	Hierarchical (Nested)	APIs and web data	Highly flexible, support complex nesting	Large file sizes, slow parsing for big data
Parquet	Columnar (Binary)	Big data analytics and storage	Massive compression, speeds up queries	Not human-readable, high compute to write
Avro	Row-oriented (Binary)	Heavy write/streaming data	Fast write speeds, excellent schema evolution	Difficult to query directly, not human-readable

11

⑪ AI is a field in computer science that builds software and systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, problem-solving and visual perception.

Narrow AI - is capable of limited to a specific, specialised domain, but fail or cannot function outside its specific programming, and it is fully operational and widely used today.

General AI - is capable of learning, reasoning and applying knowledge across any domain, and it is flexible adapts to completely new, unfamiliar situations

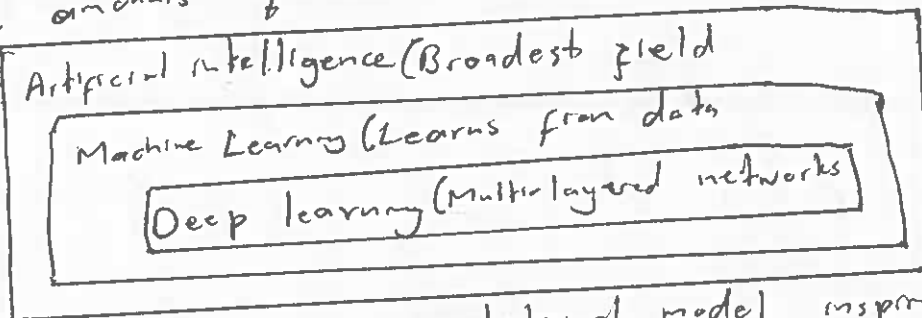
example of narrow AI is Netflix or Amazon
example of general AI is ChatGPT

⑫ Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data to improve their performance over time without being explicitly programmed.

Difference between traditional programming and ML

Aspect	Traditional Programming	Machine Learning
Input	Data + Human written rules	Data + Historical answers/outcomes
Output	The computer computes the answers.	The computer uncovers the rules/patterns.
Analogy	Following a cooking recipe exactly	Eating 50 different meals to figure out the recipe

⑬ Deep learning is a specialised subfield of machine learning that uses multi-layered artificial neural networks to learn from huge amounts of data



AI is the broad umbrella. ML sits inside AI, and Deep learning is a deeper, highly specialised subfield inside Machine Learning.

Neural network is a computational model inspired by the structure and function of biological neurons. While human brain has biological neurons, deep learning uses software-based mathematical layers to process information.

deep neural network consists of three main parts:

Input layer - receives the raw data.

Hidden layer - multiple interconnected layers that sit between the input and output. These layers automatically extract features, moving from simple shapes to highly complex concepts.

Output layer - delivers the final prediction or classification.

14) Natural Language Processing - it is a branch of AI that gives computers the ability to understand, interpret and generate human language.

- Smart Assistants (voice recognition & intent analysis) example Apple Siri, Google assistant and Amazon Alexa
- Email filtering and predictive text (auto-correct and smart reply) found in email (Gmail)
- Machine translation (Cross-Language communication) Google translate

15) Large Language Model (LLM) is a type of deep learning model trained on massive amounts of text data to understand, summarize, generate and predict human language. LLM - can perform dozens of distinct tasks (writing, coding, translating, summarizing) without needing separate models. Standard machine learning model is narrow and single-task built for one specific job.

Examples

- 1) ChatGPT - used to create, drafting professional emails
- 2) Gemini - used for heavily deep document analysis (like summarizing an entire textbook)

16) Generative AI is capable of creating entirely new, original content such as text, image, music, software code or video by learning patterns from existing human-created data. At high level it can predict and generate, it can recognise patterns.

- ChatGPT - it develops a conversational assistant that generates high quality text and code.
- Midjourney - it generates the descriptive text prompts into high realistic photos, digital painting, logos or surreal artwork
- GitHub - acts as an AI pair programmer inside code editors.

- 17) Training dataset is the initial collection of data such as text, images, numbers or audio files used to teach an AI or ML model how to identify patterns and make decisions.
- High-quality training data ensure that AI learns the correct patterns. If the training data is clean, accurate and diverse, the model will make highly precise and reliable predictions.
- If training data could be biased there would be range of software glitches to severe real-world harm such as:
- Perpetuating real-world bias and discrimination, Model hallucinations and incorrect outputs and overfitting
- 18) Computer vision is a field of AI that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs.
- Machine see and interpret image by converting pixels to numbers, feature extraction (finding patterns), building complexity, classification and action.
- examples: - Autonomous vehicles (self-driving cars)
- Medical Imaging and diagnostics (X-rays, MRIs and CT scans)
- 19) API is a software intermediary that allows two different applications to talk to each other.
- Me as a client in a restaurant sitting down looking at the menu of choice, I know what I want to eat but I cannot just go to the kitchen to cook. The kitchen is the Server/Database with raw materials/ingredients, chef and systems required to prepare my food, then waiter (the API) takes my order, walk to the kitchen, explain to the chef what I need then return to me with my food (data)
- API are important to enable connectivity and integration, saves time and money from developing or rebuilding complex tools from scratch, they enhance security, and powers data pipelines and automation
- 20) API Key is a unique, alphanumeric string of code used by a software program to identify and authenticate an application or project making a request to a server.
- Its purpose is to authenticate, use tracking and billing and rate limiting.

API key must be kept strictly confidential. Developers use several standard security practices to hide them:

- Environment variables
- The .gitignore file
- Secret managers
- API restrictions

If API key is exposed to public, massive financial losses due to hackers, data breaches will occur, service suspensions.

(21) REST API is an architectural style for designing networked applications that relies on a stateless, client-server communication protocol almost always HTTP.

REST - Representational State Transfer

GET - retrieve data (Read) - viewing a product page or looking at your shopping cart.

POST - create new data - submitting your credit card info to place brand new order.

PUT - update/replace - overwriting your entire profile address with a new one.

DELETE - remove data - clearing a product from your cart.

(22) JSON (JavaScript Object Notation) - is a lightweight, text-based data format used to store and exchange information.

It is commonly used because it is human-readable and writable, lightweight and fast, easy for computers to parse and supports hierarchical, nested structures.

Simple JSON Object

```
{  "user_id": 107835,
  "name": "Molefi Nhlapa",
  "account_balance": 1507.85,
  "email": "leroy.nhlapa@gmail.com",
}
```

23 Prompt Engineering is the process of structuring, refining and designing text inputs (prompts) to guide Generative AI models into producing the most accurate, relevant, and high-quality output possible. The skill is important to reduce errors, save time and compute, unlocks advanced capabilities, tailors tone and format.

① Poor prompt = "Write an email to my boss asking for a raise"

Improved prompt = "As an experienced corporate manager, draft a formal email to my manager, requesting a performance and salary review. Keep the tone confident, polite and professional. The email should be under 200 words."

② Poor prompt = "Explain how blockchain works."

Improved prompt = "Explain the core concept of blockchain technology. Use a simple analogy involving a public ledger notebook, and do not use complex technical terms like 'cryptographic hashing' or 'nodes'. Write it so that a 10 year old child could easily understand it. Keep it under three short paragraphs."

24 MAP function is a built-in programming tool that applies a specific function or operation to every single item inside an iterable collection and returns a brand new collection containing the transformed results.

MAP function is a fundamental operational building block of element-wise transformation - it is used to clean or reformat columns of raw data.

• foundation of map reduce - is the first phase of distributed data processing.

Python

```
prices = [10.00, 20.00, 50.00, 100.00]
```

```
def add_tax(price):  
    return price * 1.10
```

```
taxed_prices = list(map(add_tax, prices))
```

```
print ("Original Price:", prices)
```

```
print ("Taxed Price:", taxed_prices)
```

25) Cloud computing is the on demand delivery of computing services including servers, storage, databases, networking, software and analytics over the internet "the cloud"

The Three Main Cloud Service Models

Model	What it stands for	Description	Responsibility balance
IaaS e.g. Amazon	Infrastructure as a service	Rents raw computing infrastructure like servers, virtual machines and storage.	You manage the operating system, app and data; the provider manages the hardware.
PaaS e.g. Google App Engine	Platform as a service	Provides a hardware and software framework for building and deploying applications.	You only manage the code and data, the provider handles servers, OS and scaling
SaaS e.g. Microsoft 365	Software as a service	Delivers fully functional, end-user software applications over the web.	The provider manages everything, you simply log in and use the software

26) Version Control System (VCS) is a software tool that tracks and manages changes made to source code or data scripts overtime.

Git is a free, open-source distributed version control system. Analysts use it because it has safe collaboration (allows members to work on the exact same codebase or analysis scripts simultaneously without accidentally overwriting each other's work), error recovery (accidentally deletes a database column or breaks a dashboard, Git allows them to instantly undo the error and restore last working version), Audit trail (transparent, permanent ledger of every single modification, making it easy to track how a predictive model or data pipeline has evolved over months or years).

27) Jupyter Notebook is an open-source, interactive web application that allows users to create and share documents containing live, executable code, equations, visualisations and explanatory narrative text. It is commonly used with Python. Analysts and Data Scientists use it for interactive data exploration, inline data visualisations, iterate programming and storytelling, and rapid prototyping for machine Learning.

(28) Python is a high-level, interpreted and general purpose programming language known for its extreme readability, simpler syntax and versatility.

Python has become the undisputed industry standard for data work due to:

- Syntax like English
- A massive, specialised ecosystem
- Exceptional community and industry support
- Seamless integration

Python Libraries

Numpy (Numerical Python) - it introduces a powerful multi-dimensional array object that processes massive mathematical computations dramatically faster than standard Python lists.

Pandas - It is built on top of Numpy, it is fast, flexible data structures called DataFrames, which mimic structured Excel spreadsheet or SQL data tables inside the code.

Matplotlib - It is a comprehensive, low-level data visualisation library capable of creating static, animated and interactive plots.

Scikit-Learn - is the industry-standard machine learning library of traditional data science. It features a clean, highly consistent interface for implementing dozens of statistical algorithms.

(29) Data Privacy - it focuses on proper handling of personal information - covers names, emails, financial records and ID numbers. - grants individual control over their digital footprint.

It is important to prevent identity theft and financial fraud. It protects personal reputation and safety.

It builds consumer trust in digital services.

It stops unauthorised corporate tracking and manipulation.

GDPR is EU's strict data privacy law, it gives users rights to access and delete their data, it imposes massive fines to non-compliance.

It applies to all 27 European Union member states, European Economic Area countries: Iceland, Liechtenstein and Norway UK, and any company world wide if it processes data of individuals located in the EU/EEA.

30) Data ethics is the study of how data is collected, shared and used, specifically focusing on fairness, transparency and accountability.

examples

- (1) Harvesting personal data without their explicit consent on social media platform to do political advertisements.
Problematic: it is violation of consent and democratic subversion.
- (2) Algorithmic bias to predict in the court the likelihood black person will reoffend, against the white person.
Problematic: it is perpetuating a systemic racism.
- (3) Non-consensual AI Image generation that create images/video of an artist to mimic exact style to replace their livelihood.
Problematic: it cause reputational harm, affects psychological and economic exploitation to the artist.

31) Data governance is the formal collection of people, processes and technologies that define how an organisation manages, protects and uses its data assets.

- (1) Strategic roles and accountability
- (2) Data Quality Management
- (3) Data Architecture and Metadata Management
- (4) Security, Privacy and Compliance
- (5) Data policies and Standards
- (6) Training and Data Culture

32) Bias in AI is systematic, unfair or prejudiced outcomes generated by an AI system that disadvantage certain groups of people, typically along the lines of race, gender, age or socioeconomic status.

The consequences of AI bias: are socioeconomic exclusion, discrimination in healthcare, unjust legal outcomes and erosion of public trust.

- 33
- Data Analyst looks at the past to explain what happened.
 - Data Engineer builds the infrastructure that allows data to flow securely.
 - Data Scientist looks at the future to predict what will happen
 - AI Engineer builds intelligent systems that can reason, see or chat

Feature	Data Analyst	Data Engineer	Data Scientist	AI Engineer
Core Focus	Interpreting historical data to provide actionable business insights	Building pipelines and infrastructure to store and move data.	Creating predictive models and uncovering hidden patterns.	Deploying, fine-tuning and integrating machine learning and LLM systems into software.
Primary Question	"What happened, and why did our revenue change last quarter?"	"How do we securely move millions of data points per second into our database?"	"Can we build a model to predict which customers are likely to cancel next month?"	"How do we integrate a conversational AI chatbot into our existing app architecture?"
Key skills	Data storytelling, dashboard creation, basic statistics, business intelligence	ETL pipeline building, data warehouse, system architecture, database optimization	Advanced statistics, machine learning, experimental design, predictive modeling	Deep learning, NLP, APIs integration, prompt engineering, MLOps (Machine Learning Operations)
Primary Tools	SQL, Excel, Tableau, Power BI, Google analytics	SQL, Python, Apache Spark, Docker, Airflow, AWS/Azure, Snowflake	Python, R, SQL, Pandas, Scikit-Learn, Jupyter, Notebooks	Python, PyTorch, TensorFlow, Hugging Face, OpenAI APIs, LangChain, Kubernetes

34) Business Intelligence (BI) is the procedural and technical infrastructure that collects, stores and analyses the data produced by a company activities. It simplifies raw, complex data into easy-to-understand reports, dashboards and graphs so leaders can make informed, data driven decisions

Power BI - a cloud-based business analytics service that integrates seamlessly with the Microsoft ecosystem. It connects to hundreds of data source simultaneously to build real-time interactive dashboards

Tableau - highly visual, enterprise-grade data visualisation platform celebrated for its drag-and-drop simplicity and handling of datasets. It allows users to ask questions of their data using natural language and convert to visual charts.

(35) The interesting concept is of deep learning, I was not aware how AI gets to be functional as it is, so now I know that there are deeper fields that are involved prior to AI effectiveness, and how neural network is computerized into processing huge amounts of data and the bridging models of it to enable machine learning to learn from the data provided to handle better outcomes in the future.

(36) Cloud computing models. I have to do more research as I found a table that gives descriptive and distinction between IaaS, PaaS and SaaS.

(37) Everything is getting digital, so data and AI will affect my career by analyzing patients results without me since it can even refer to the textbooks, case-studies and historical reports to determine the outcome. Example for patients with Malignancy, drawing a flow cytometry test to analyze the antibodies data and AI will do that task, give report, and suggest the possible outcome, and refer for further tests on that patient specimen.



BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals



Research Tip: Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

Database is where data/information of an organisation/institution is stored electronically in a computer system

BrightLearn Data Analytics | Page 1 of 13

Essential because it can be fast to retrieve data, for data security, allows many people to view and update information at exact same time without crashing or mixing up data, it built-in rules to make sure information stays accurate and reliable across all systems.



- Banking and finance sectors - banks use DBs to record accounts balances, process ATM withdrawals and log money transfers instantly and securely
- Health sector
- Sale Online retail - e.g. Takealot - use dbs to store product prices, track real-time stock levels and manage user orders.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse? Data warehouse is a centralized computer system that collects, integrates and stores large volume of data whether both current and historical from multiple database systems.

It differs from regular db primarily its purpose of analytical processing - designed for complex queries, forecasting and trend analysis. It uses online analytical processing, it deals with historical data in the system.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.



6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.



10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.



Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.



15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.



23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.



27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.



Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.



33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?



37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. www.scibbr.com / How to cite a website in APA style / 2020
2. www.studocu.com // Cyber . 2023 . Data Governance Frameworks.
3. Smith J (2022) How to build a Model AI . IBM Think
4. European Commission . 2018 . General Data Protection Regulation Key . GDPR.eu . gdp.eu
5. ProPublica . 2016 . Machine Bias . propublica.com . UN World . org
6. Gartner . 2023 . Gartner Glossary: Data Governance . gartner.com
7. Microsoft . 2026 . What is Power BI . Microsoft Learn . microsoft.com
8. Tableau . 2026 . What is Business Intelligence? . Tableau . tableau.com

Good luck! We believe in you.

BrightLearn Data Analytics